Unit Test Report

Automated unit test execution and outcome

Event Management System (Eventify)

1. Objective

This report records automated frontend and backend unit-test execution for the Eventify project. It summarizes test counts, outcomes, execution commands and known test-run caveats.

2. Test environment and commands

- Operating system: Ubuntu local development environment.
- Runtime/tooling: Node.js v22.22.1 and npm 9.2.0.
- Frontend: React client test suite executed from the client directory using `npm test`.
- Backend: Node/Express server test suite executed from the server directory using `npm test`.
- Coverage rerun: backend `npm test -- --coverage`; frontend `npm test -- --experimental-test-coverage`.

3. Unit test results

Test area	Executed command	Passed	Skipped	Failed	Outcome
Frontend unit tests	client: npm test	3	0	0	PASS
Backend unit tests	server: npm test	289	6	0	PASS with skipped tests
Backend test suites	Jest suite summary	21 suites	1 suite	0 failing suites	PASS with skipped suite

Backend Jest summary: 21 passed suites and 1 skipped suite (22 total); 289 passed tests and 6 skipped tests (295 total); 0 failed. Runtime reported approximately 9.7 seconds in the standard run.

Frontend summary: 3 tests passed and 0 failed. The tests cover registration history state, seat-count synchronization during register/cancel operations, and mapping backend error status/messages into the UI.

4. Key automated test coverage areas

- Frontend registration-state behavior: active registration history, registration/cancellation state transitions and seat-count synchronization.
- Frontend registration error handling: uses backend status and message contracts.
- Backend authentication and authorization, event and registration controllers, input validation, password security, secret leakage safeguards, audit logging, notifications, models, routes and related utilities.
- Backend regression suite includes registration behavior and error/negative-path checks; six tests and one suite were skipped and are not counted as passes.

5. Supporting quality checks

- Client ESLint completed with no reported lint errors.
- Client production build completed successfully using Vite; 36 modules were transformed and build completed in approximately 296 ms.
- Backend test run emitted expected console warnings/errors from exercised negative scenarios. These did not result in failed tests.

6. Observations and follow-up

- Review the six skipped backend tests and the skipped suite; document why they are skipped and decide whether they should be enabled for the final regression run.
- A Mongoose deprecation warning was observed for `Document.prototype.validateSync()`; review compatibility ahead of Mongoose 10.
- Application logs also reported a deprecation warning recommending `returnDocument: 'after'` instead of the `new` option in `findOneAndUpdate()` / `findOneAndReplace()`.

7. Conclusion

The executed unit-test suites completed with zero failures. The backend suite contains skipped tests, so the result should be reported as a passing executed suite with skips disclosed, not as complete execution of every test in the repository.